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ExpressNet-MoE: A hybrid deep neural network for emotion recognition☆

Deeptimaan Banerjee, Prateek Gothwal, Ashis Kumer Biswas *

Computer Science and Engineering, University of Colorado Denver, Denver, 80204, CO, USA

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ABSTRACT

In many domains, including online education, healthcare, security, and human–computer interaction, facial emotion recognition (FER) is essential. Real-world FER is still difficult because of factors like head positions, occlusions, illumination shifts, and demographic diversity. Engagement detection system, which is essential in virtual learning platforms is severely challenged by these factors. In this article, we propose ExpressNet-MoE, a novel hybrid deep learning architecture that combines Convolutional Neural Networks (CNNs) with a Mixture of Experts (MoE) framework to address these challenges. The proposed model dynamically selects the most relevant expert networks for each input, thereby improving generalization and adaptability across diverse datasets. Our methodology involves training ExpressNet-MoE independently on several benchmark datasets after preprocessing facial pictures using BlazeFace for face detection and alignment. To maintain class distribution, stratified sampling is used to divide each dataset into training and testing groups. Our model improves on the accuracy of emotion recognition by utilizing multi-scale feature extraction to collect both global and local facial features. ExpressNet-MoE includes numerous CNN-based feature extractors, a MoE module for adaptive feature selection, and finally a residual network backbone for deep feature learning. To demonstrate efficacy of our proposed model we evaluated it on four widely used datasets: AffectNet_7, AffectNet_8, RAF-DB, and FER-2013; and compared with current state-of-the-art methods. Our model achieves accuracies of 74.40% ± 0.45 on AffectNet_7, 71.98% ± 0.66 on AffectNet_8, 83.41% ± 1.06 on RAF-DB, and 67.05% ± 2.08 on FER-2013. Overall, the findings indicate that the adaptive expert selection and multi-scale feature extraction significantly enhances the robustness of facial emotion recognition across diverse real-world conditions and how it may be used to develop end-to-end emotion recognition systems in practical settings. Reproducible codes and results are made publicly accessible at https://github.com/DeeptimaanB/ExpressNet-MoE.

1. Introduction

Facial Emotion Recognition (FER) has become a major component of virtual education (Farman et al., 2023), security (Ghadekar et al., 2024), medical industry (Bohi et al., 2024; Huang, Wu, et al., 2023), and human–computer interfaces. The ability to sense and respond to human emotions has created new opportunities for developing more flexible and intelligent systems. However, since human facial expressions are inherently complex, FER in real-world scenarios is a difficult task. Variations in head position, occlusion, lighting conditions (Farman et al., 2023; Sajjad et al., 2023; Zhang et al., 2023) and demographic variability frequently cause significant performance issues in emotion detection frameworks.

The performance of FER frameworks has improved significantly over time. There have been multiple ways for FER and its classification, including Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG), combined with traditional machine learning algorithms such as Support Vector Machines (SVM) (Shan et al., 2005, 2009) and Random Forests (Nazir et al., 2018; Salhi et al., 2012). Convolution Neural Networks (CNNs) have played a pivotal role in learning hierarchical feature representations (Khodaverdian et al., 2021). Moreover, transfer learning-based models have further enhanced recognition of complex emotional expressions due to the fact that they are pre-trained on multiple datasets. However, there are still many challenges in the machine learning development pipeline, such as unbalanced datasets and intra-class variations, leading to limited generalization.

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* Corresponding author.
E-mail addresses: deeptimaan.banerjee@ucdenver.edu (D. Banerjee), prateek.gothwal@ucdenver.edu (P. Gothwal), ashis.biswas@ucdenver.edu (A.K. Biswas).
URLs: https://ml.cse.ucdenver.edu/people/deeptimaan_banerjee (D. Banerjee), https://ml.cse.ucdenver.edu/people/prateek_gothwal/ (P. Gothwal), https://cse.ucdenver.edu/~biswasa (A.K. Biswas).

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Hence, we propose ExpressNet-MoE, a novel hybrid Deep CNN architecture. It is a hybrid deep learning model that combines CNNs with a layer of mixture of experts (MoE) to overcome the aforementioned problems. This method dynamically selects the relevant expert networks for each input, while improving its versatility and generalization over a number of datasets. ExpressNet-MoE captures simultaneously both global and fine-grained facial features and improves the accuracy of the emotion recognition task. The hybrid architecture of the model offers the most adaptive feature selection capability, and the choice of multiple CNN-based feature extractors makes it a powerful yet flexible approach.

The contributions of our research include the following.

- **Adaptive feature learning**: Many existing solutions employ static models offering only a fixed set of features. To overcome the limitations of static feature extraction models, ExpressNet-MoE chooses a Mixture of Experts (MoE) system that emphasizes the best expert network in the model for each individual input.
- **Multi-scale feature extraction**: Our model extracts both global and fine-grained facial expression characteristics using CNNs with different filter sizes. The model's capacity to identify minute emotional variations across datasets is enhanced by the choice of this hybrid method.
- **Improved generalization across multiple datasets and real-world scenarios**: Our testing results showed that the model improves the generalization capability across different datasets due to its hybrid architecture. This allows the model to handle real-world issues such as illumination, occlusion, and demographic diversity.

We have evaluated ExpressNet-MoE on three benchmark FER datasets which are AffectNet (Mollahosseini et al., 2017), Real-world Affective Faces Database or RAF-DB (Li & Deng, 2019; Li et al., 2017), and FER-2013 (Goodfellow et al., 2013). Each of these datasets has their own set of unique characteristics and problems. The AffectNet dataset is well-suited for training and validation since it provides a vast and varied collection of facial emotions from real-world situations. RAF-DB is reputed for its precise labeling; therefore, it is an ideal choice for testing the model's generalizability. FER-2013 is a highly appropriate dataset to test the model's performance under constrained circumstances as it contains grayscale images with poor resolutions and possible mislabeling issues. We evaluate our model on these datasets to show how it responds to variations in facial expressions, quality of images, environmental factors, and demographics.

Furthermore, ExpressNet-MoE employs deep feature representations, which offers more data-driven manner due to its dynamic MoE and transfer learning architecture to comprehend user emotions than conventional fixed CNN-based solutions.

The remainder of this article is organized as follows: Section 2 reviews related works related to current state-of-the-art emotion recognition methods. Section 3 provides the datasets utilized for both training and evaluation of our proposed model, along with comparing its performance with existing work. In Sections 4 and 5, we present our proposed methodology and ExpressNet-MoE architecture, and its associated machine learning pipeline needed for end-to-end learning. Section 6 illustrates the experimental results of the proposed method separately for each of the datasets considered in this study. A comparative analysis is described in Section 7. Section 8 provides an insight on the model performance and efficacy considerations. Finally, Section 9 summarizes the study, offers our conclusion and elaborates on future work.

2. Related works

Understanding facial emotions has become a key element in emotion detection frameworks, particularly in applications such as human–computer interaction and online learning. Prior research has extensively investigated improvements in FER model accuracy, scalability, and generalization under real-world conditions. Several recent studies proposed architectures that combined attention mechanisms with convolutional feature extraction. The Dual-Direction Attention Mixed Feature Network (DDAN-MFN) integrated a Mixed Feature Network with a Dual-Direction Attention Network to enhance long-range dependency modeling (Zhang et al., 2023). The MFN used convolution and bottleneck layers along with MixConv, which leveraged differing kernel sizes. The DDAN had an independent dual-direction attention head to capture long-range dependence, which significantly enhanced the ability of the model to highlight the features of the FER undertaking.

Similarly, The Generalized Zero-Shot Convolution Neural Network (GZS-ConvNet) addressed the generalization problem in FER systems (Bhati et al., 2025). The architecture was designed to detect unseen facial expressions using an adaptation mechanism. Due to its ability to perform zero-shot categorization, GZS-ConvNet demonstrated high performance on a variety of datasets including FER-2013, AffectNet, and RAF-DB, highlighting its suitability for dynamic real-world environments. Similarly, the ConvNeXt, a CNN architecture, demonstrated superiority of its FER performance over classical CNN and Transformer-based systems (Bohi et al., 2024). The architecture used deep convolutions, depthwise convolutions, and a modified activation function. The model showed competitive performance on the AffectNet dataset. Face2Nodes, a graph-based FER system used dynamic relation-aware graph convolutions to characterize spatial and relational relationships between face regions (Jiang et al., 2023). It, thus, allowed the system for more structured and expressive emotion representations. Their work demonstrated strong performance on the RAF-DB dataset.

There are few demonstrated efforts exploring techniques to avoid overfitting and generalization problems: (i) The work on deep convolution layers for emotion categorization using strategies such as max pooling, dropout, and batch normalization (Uniyal & Agarwal, 2024). Their method underlined the importance of feature extraction and regularization to achieve high accuracy on large datasets identical to AffectNet and FER-2013. (ii) The use of ensemble modeling with VGG and ResNet for emotion and engagement in distance learning (Savchenko et al., 2022).

Multimodal data and attention mechanisms have been successfully integrated to improve FER systems. In order to enhance emotion recognition, a multi-modal sentimental privileged information embedding (IA-MTM) was introduced that integrated audio and picture characteristics (Sun et al., 2024). Using both visual and aural input, their model improved FER performance. The model employed the ResNet18 architecture for image feature extraction and an audio-decoding network to create a shared feature space. The Cross Similarity Attention (CSA) mechanism was introduced to address class imbalance by explicitly increasing inter-class separation while reinforcing intra-class similarity (Wang et al., 2024).

A CNN-based architecture for FER effectively classified facial expressions into positive, negative, and neutral classes using global average pooling and residual depth-wise separable convolutions (Rajavenkatanarayanan et al., 2018). In order to improve feature extraction and learning stability, ResEmoteNet, combined residual networks, convolution blocks, and Squeeze and Excitation (SE) blocks (Roy et al., 2024). Their model's capability to mitigate the impacts of vanishing and exploding gradients was demonstrated by its strong performance on a number of datasets.

The SE-ResNet architecture combined Squeeze-and-Excitation with the ResNet architecture (Huang, Chiang, et al., 2023). The model captured the importance of each channel as it integrated the SE block into the ResNet architecture. It demonstrated an improved learning performance during training and increased accuracy in FER tasks.

On the other hand, the classical machine learning algorithms, such as the Bayesian classifiers and C4.5 trees were also investigated in the literature to understand affective states and involvement in online learning environments (Dewan et al., 2019). These algorithms analyzed

motions and facial expressions to help improve the tracking of student attention.

The Deep Convolution Neural Network with Convolution Block Attention Module (DCNN-CBAM), made use of the attention mechanism which improved the feature representation (Halim et al., 2023). This design, which used deep layers and convolution blocks was tailored for the AffectNet_8 dataset, showed promise in identifying students' emotional states during online instruction.

Transformer-based architectures have also made their way into FER because of their capacity to recognize long-range correlations in face characteristics. For emotion classification, a Vision Transformer (ViT) model was pre-trained on the ImageNet-21k and further fine-tuned on the AffectNet dataset (Roka & Rawat, 2023). The findings demonstrated that leveraging large-scale datasets alongside data augmentation strategies is vital for managing class imbalance in FER tasks. Similarly, the hybrid framework, FER-VT, combined CNNs with attention mechanisms (Huang et al., 2021). It introduced Visual Transformer Attention (VTA) for high-level semantic representation and Grid-Wise Attention (GWA) for low-level feature extraction.

Both studies demonstrated a growing trend toward transformer-based approaches in facial emotion recognition systems. These techniques improved upon traditional CNN architectures by enabling more accurate representation of long-range spatial dependencies in facial imagery. By integrating attention mechanisms across multiple feature levels, transformer-based models demonstrated an enhanced ability to represent both local and global facial cues. Consequently, these architectures gained increasing adoption for FER in complex real-world environments.

FER systems have advanced significantly as outlined above. However, there are still a number of research gaps that prevent their practical applications. Due to variances in demography, lighting, and face features, many current models have trouble generalizing across datasets; they perform well on certain datasets but fall short when evaluated on unseen data (Huang, Chiang, et al., 2023; Wang et al., 2024; Zhang et al., 2023). Conventional CNN-based FER models are limited in their capacity to adapt to a variety of inputs because they rely on set feature extractors, which might not be the best for all facial expressions (Huang, Chiang, et al., 2023; Uniyal & Agarwal, 2024). Furthermore, it is challenging to capture both global face structures and fine-grained expression features since the majority of architectures do not integrate multi-scale feature extraction (Huang, Chiang, et al., 2023; Uniyal & Agarwal, 2024; Wang et al., 2024). Managing occlusions, light fluctuation, and non-frontal head poses—all of which impair model performance, this is another crucial issue in real-world FER.

Dataset imbalances also make it difficult to classify underrepresented emotions like disgust, fear, and contempt, restricting the model's capacity to confidently identify uncommon expressions, which is evident in Bohi et al. (2024), Wang et al. (2024), Zhang et al. (2023). Low-resolution grayscale images and mislabeling problems in the popular dataset FER-2013 pose additional difficulties that can have a big influence on model learning and generalization. By utilizing a Mixture of Experts (MoE) framework for adaptive feature selection, incorporating multiple CNN-based extractors for multi-scale learning, and improving generalization across datasets by combining deep feature learning and ensemble-based decision-making, ExpressNet-MoE directly addresses these limitations. ExpressNet-MoE is a very flexible and scalable solution for real-world FER applications since it enhances robustness against occlusions, illumination shifts, and dataset bias by dynamically choosing expert networks based on input data.

3. Datasets

We have used three benchmark datasets which are AffectNet, RAF-DB, and FER-2013 to train our model. Each of the aforementioned datasets have distinct qualities that will help improve robustness and generalizability of the model. AffectNet is perfect for learning complicated emotional variations since it offers a large collection of face expressions that have been taken in real-world situations. RAF-DB's extensive collection of extremely diverse photos with well-documented annotations will help the model to test its adaptability and generalizability. Furthermore, the grayscale photographs in FER-2013 were taken in a variety of settings to help the model adapt to variations in lighting and facial expressions.

In order to balance computational efficiency, and memory constraints, we employed subsets from AffectNet and RAF-DB. Fig. 1 illustrates the total number of images for every dataset per emotion class along with their training and testing splits along with the class distributions. AffectNet's 8-class classification is denoted by AffectNet_8, whereas AffectNet's 7-class classification is denoted by AffectNet_7. We increased the model's adaptability by utilizing these datasets making the model more robust.

3.1. AffectNet

AffectNet is one of the biggest datasets for facial expression identification (Mollahosseini et al., 2017). It contains over a million photos and about 440,000 samples have annotations. We have chosen as subset of this dataset with 28,175 photos from eight different emotional categories from the mentioned dataset for this study which are surprise, happy, anger, disgust, neutral, fear, sad, and contempt. These photos are collected from online sources and are taken from actual real-life situations and contain a variety of facial expressions. Because of its vast size and computational constraints, a subset of this dataset is generally used by many researchers in their works (Bhati et al., 2025; Savchenko et al., 2022; Uniyal & Agarwal, 2024).

For training, we have used 22,540 images, and 5635 were set aside for testing. The selected dataset includes a wide range of age groups, ethnicity, and lighting conditions, making it ideal for improving generalization and the model's adaptability. To make a 7-class classification on the AffectNet dataset, we remove the Contempt category and made a training set of 19,944 images and a testing set of 4987. Although AffectNet offers a wide variety of emotions, the distribution of classes is imbalanced compared to the original which is heavily imbalanced, with Happy and Surprise being more represented than Neutral and Fear.

Because of its versatility, AffectNet is a great tool for training models to function well in multiple settings. The excellent careful annotations in the dataset ensure that the model learns from precise labels and aid in improving emotion classification. The dataset is a popular option for emotion detection despite its class imbalance.

3.2. RAF-DB

Another popular dataset for facial emotion detection is RAF-DB (Li & Deng, 2019; Li et al., 2017). It consists of 15,339 single class photos tagged with the seven main emotions classes that are surprise, fear, happiness, anger, disgust, neutrality, and sadness. It contains pictures of people of all ages, genders, and cultural backgrounds. It stands out for its diversity. This variation in the dataset makes it effective for training models that must function well across various populations.

We used 3068 photos for testing and 12,271 images for training from the RAF-DB dataset. The photographs in this dataset like AffectNet were also taken in real-world settings and are downloaded from the internet. It contains variations in head positions, backgrounds, and lighting. Building a strong model that can identify emotions outside of controlled environments requires an understanding of these real-world difficulties. There is a noticeable class disparity in RAF-DB, as happy and neutral classes are significantly more prevalent than fear, anger and disgust.

We proved the model's generalizability by identifying nuanced emotional features by using RAF-DB dataset. The dataset complements the AffectNet and FER-2013 datasets, as the dataset guarantees a more comprehensive portrayal of emotions. The reliability of the emotion classification model is further increased by the high-quality training data provided by its well-annotated images.

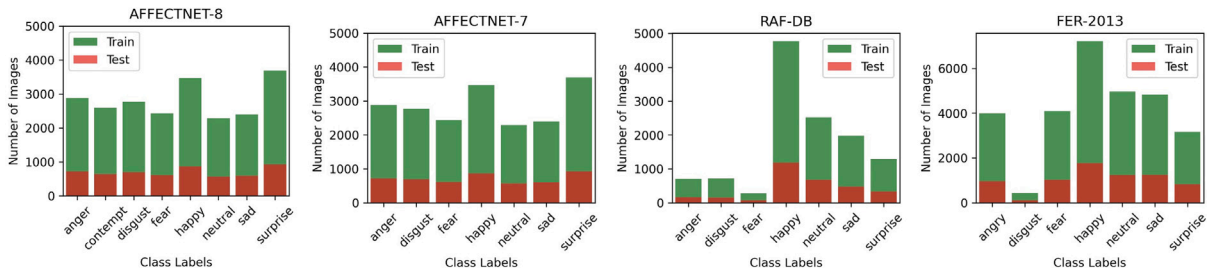

Fig. 1. Training vs Test splits for the datasets.

3.3. FER-2013

FER-2013, which was first presented in the ICML 2013 challenge (Goodfellow et al., 2013), is a well-known dataset in facial expression recognition. There are 35,887 grayscale photographs in the dataset, each with a resolution of 48 by 48 pixels. It contains 7 different emotion classes which are anger, disgust, fear, happiness, neutrality, sadness, and surprise.

Although FER-2013 is widely used, it has a number of drawbacks. The first one is mislabeling (Barsoum et al., 2016; Mazen et al., 2021), wherein some images are mistakenly categorized as belonging to the wrong class of emotion. This may impact the model's performance as it introduces noise during training. Furthermore, the low-resolution photos in the dataset makes it more difficult for models to extract fine-grained facial traits. On top of that, emotions like happy and neutral have far more samples than other classes like disgust and surprise, indicating a class imbalance. The model may become less sensitive to underrepresented emotions as a result of this imbalance, which could result in biased predictions. These drawbacks are somewhat offset by its size and variety of real-world examples in the classes with more samples. For this study, we utilized 28,709 images for training and 7178 for testing.

4. Proposed methodology: ExpressNet-MoE

The proposed methodology involves training the model on three facial expression recognition datasets: AffectNet, RAF-DB, and FER-2013 (individually, to test generalizability). Two variations of AffectNet were taken into consideration: one with seven emotion classes with the "contempt" category eliminated - AffectNet$_7$, and another with all the eight classes kept - AffectNet$_8$. There are seven emotion classes in RAF-DB as in AffectNet$_7$. To maintain compatibility with other datasets for the model's input, the grayscale images in FER-2013 were transformed into three channels. To guarantee an equitable distribution of emotion categories in both training and testing sets, a stratified train–test split is used for all datasets. This keeps the model's performance from being distorted by data imbalance as much as possible. We use the same test split for evaluating our model as well as comparing our proposed model with the state-of-the-art methods.

The BlazeFace model (Bazarevsky et al., 2019) from MediaPipe, which effectively identifies facial areas in photos, is used for face identification and alignment. It is used by many researchers for preprocessing facial images (Bayar et al., 2022; Hossen & Uddin, 2023; Soudy et al., 2024). The identified faces are cropped after the relative bounding-box coordinates are calculated and gently adjusted for robustness. The original image is scaled to 224 × 224 pixels if no face is found. An image generator that dynamically loads photos, does preprocessing (using BlazeFace), normalizes pixel values, and one-hot encodes emotion labels is included into the data pipeline. To maximize training efficiency and avoid overfitting, the model is trained for 15 epochs with a batch size of 32 using methods such as checkpointing, learning rate adjustments, and early stopping. This methodology ensures effective learning across diverse datasets and also improving the model's generalization for facial expression recognition. Fig. 2 shows the complete methodology.

Accuracy, precision, recall, and F_1-score are used to evaluate facial expression recognition algorithms. Accuracy measures the overall proportion of correct predictions, giving a general sense of how well the model performs. Precision focuses on how many of the predicted expressions were actually correct. Recall, on the other hand, assesses how many of the true expressions the model was able to identify. The F1-score combines precision and recall into a single value, and provides a balanced measure which is especially useful when a dataset is imbalanced. These metrics provide a thorough evaluation of the model's performance.

5. Model architecture

A number of significant hyperparameters that control the component structure and training behavior incorporated into the proposed architecture are listed in this section. These include parameters associated with the Mixture of Experts (MoE) module, such as the number of experts and gating methods, as well as kernel sizes, filter counts, dropout rates, activation functions, and normalization strategies within the CNN-based feature extractors. It also includes the hyperparameters used in training such as learning rate, batch size, optimizer, and loss function. The sections that follow offer a thorough description of these hyperparameters. We tried several different hyperparameters for the model including different filter sizes, activation functions, depth of CNN layers, etc. and finally selected parameters that worked best and still provided a reasonable model size.

5.1. CNN Feature Extractor 1

CNN Feature Extractor 1 or CNNFE1 (Fig. 3) is a Deep Convolution Neural Network (D-CNN) designed to extract hierarchical spatial information from a 224 × 224 × 3 input images.

To preserve spatial dimensions, the network begins with a 75 × 75 convolution layer with 8 filters, ReLU activation, and "same" padding. The convolution operation is defined in Eq. (1), where $Y(i,j)$ is the output feature at position (i,j), $X(i+m, j+n)$ is the input or previous layer's output, and $K(m,n)$ is the filter kernel.

$$Y(i,j) = \sum_m \sum_n X(i+m, j+n) * K(m,n) \tag{1}$$

The ReLU formula is given in Eq. (2). ReLU (Rectified Linear Unit) is a widely used activation function that outputs the input (x) if positive, or zero otherwise. It introduces non-linearity, it is computationally efficient, and helps prevent vanishing gradients. This enables deeper networks to train effectively.

$$f(x) = \max(0, x) \tag{2}$$

A dropout layer (rate = 0.1) follows to reduce overfitting, then batch normalization is applied to stabilize and accelerate training. Given

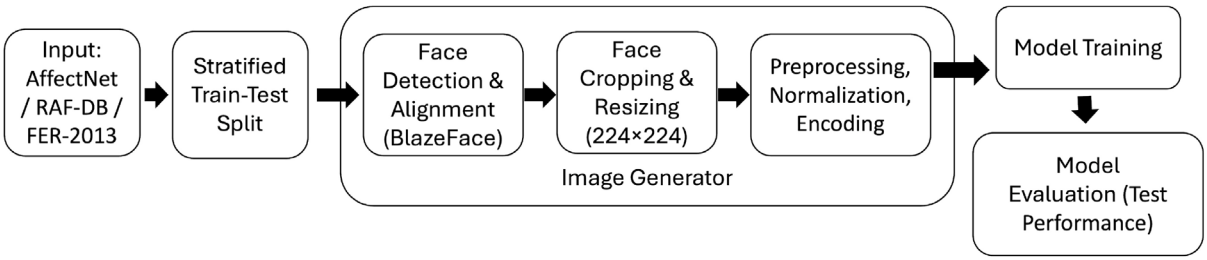

Fig. 2. Proposed methodology.

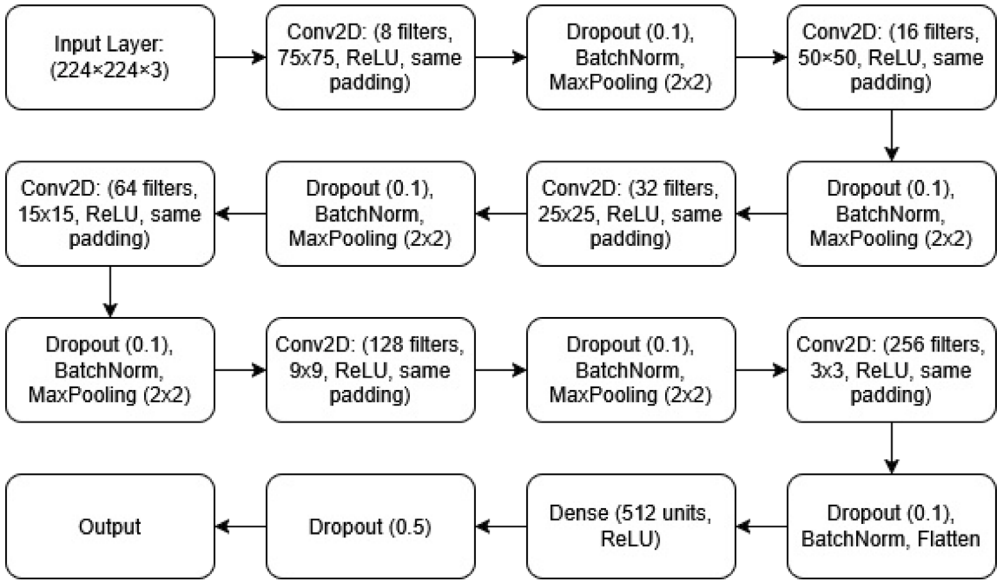

Fig. 3. CNN Feature Extractor 1 (CNNFE1).

input x_i in a mini-batch of size m, batch normalization proceeds as follows:

$$m_B = \frac{1}{m}\sum_{i=1}^{m} x_i \tag{3}$$

$$s_B^2 = \frac{1}{m}\sum_{i=1}^{m}(x_i - m_B)^2 \tag{4}$$

$$c_i = \frac{x_i - m_B}{\sqrt{s_B^2 + \epsilon}} \tag{5}$$

$$y_i = \gamma c_i + \beta \tag{6}$$

Eqs. (3) and (4) compute the batch mean m_B and variance s_B^2 for input x_i, where m is the batch size. Eq. (5) normalizes the input using the batch statistics, with ϵ preventing division by zero. Here, γ and β are learnable scale and shift parameters used in batch normalization. A 2×2 max-pooling layer then reduces spatial dimensions and improves efficiency. The max-pooling operation is defined in Eq. (7).

$$Y(i,j) = \max_{m,n} X(i+m, j+n) \tag{7}$$

Following the first convolution layer, the feature extraction process continues with progressively smaller kernels and more filters: 50×50 (16), 25×25 (32), 15×15 (64), 9×9 (128), and 3×3 (256), each followed by dropout, batch normalization, and pooling (except the last). The output is flattened and passed to a dense layer with 512 neurons (ReLU). The dense layer (i.e., fully connected layer) calculates its outputs (y) using its definition listed in Eq. (8).

$$y = f(Wx + b) \tag{8}$$

W is the weight matrix of the layer and b is the bias vector for the current layer. f is the activation function used (in this case, ReLU).

This is finally followed by dropout (0.5) to prevent overfitting. The model captures both low and high-level spatial features efficiently.

5.2. CNN Feature Extractor 2

CNN Feature Extractor 2 or CNNFE2 as shown in Fig. 4. is also a CNN architecture designed to extract hierarchical spatial features while maintaining computational efficiency.

It starts with a 15×15 convolution layer (16 filters, ReLU, same padding), followed by dropout (0.1), batch normalization, and 2×2 max pooling. This structure repeats with 32 filters (7×7), 64 filters (5×5), 128 filters (3×3), and finally 256 filters (3×3), each layer followed by batch normalization, ReLU, and pooling. Batch normalization is used to stabilize the training process and the dropout layers are applied to reduce overfitting. Instead of flattening (like in CNNFE1), a Global Average Pooling (GAP) layer reduces each feature map to a single value, creating a compact and efficient feature vector. GAP for a given feature map X of size $H \times W$ (height $\times$ width) is computed by the equation given in Eq. (9).

$$Y = \frac{1}{H \cdot W}\sum_{i=1}^{H}\sum_{i=1}^{W} X(i,j) \tag{9}$$

$X(i,j)$ is the value of the feature map at position (i,j) and Y is the scalar output representing the average value of the entire feature map.

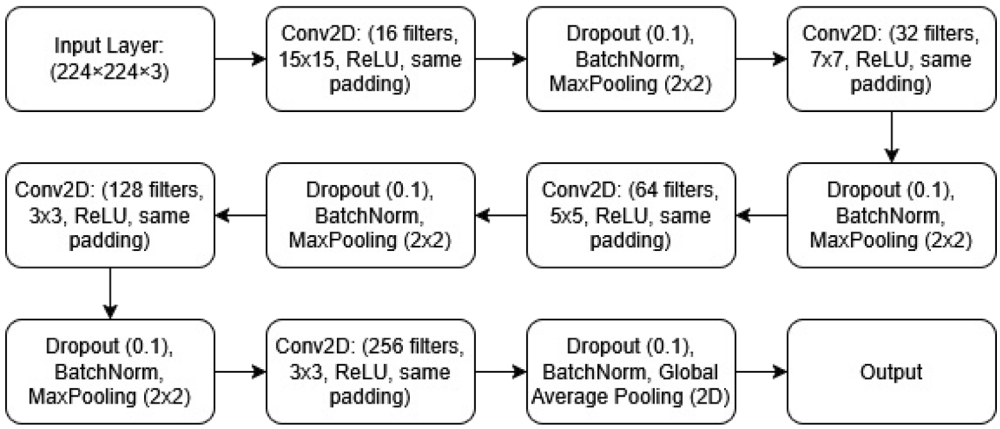

Fig. 4. CNN Feature Extractor 2 (CNNFE2).

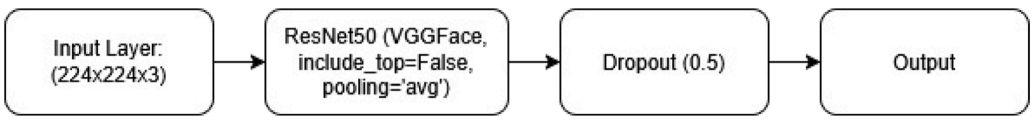

Fig. 5. ResNet-50 architecture.

5.3. ResNet-50 model

A pre-trained ResNet-50 model trained on the VGGFace2 dataset (Cao et al., 2018) originally tasked for face recognition was leveraged here to acquire the high-level face features. It ensures deep feature learning by processing 224 × 224 × 3 images through the residual blocks of ResNet50. We have used `include_top=false` that turns the model into a pure feature extractor by removing the classification levels. In order to avoid overfitting, a 0.5 dropout layer is used after global average pooling which condenses feature maps into a compact vector. Transfer learning is used in this architecture to provide reliable face expression recognition. This sub-model architecture is shown in Fig. 5.

5.4. Mixture of Experts

Our model's Mixture of Experts (MoE) module enhances the performance by dynamically choosing the most pertinent expert or experts for each input by utilizing a variety of expert networks. The structure of MoE is given in Fig. 6.

The network takes a vector of size `input_dim` as input and starts with a dense input layer. After that, this input is run in parallel through a number of expert models (denoted by `num_experts`, e.g., 4 experts), each of which has a dense layer with ReLU activation to enable it to capture distinct data features. To regulate the information flow, a gating network is implemented, with `num_experts` logits generated by its own dense layer. A Softmax activation is used to convert these logits into a probability distribution over the experts, allowing the gating mechanism to choose the final experts. The softmax is defined Eq. (10).

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}, \tag{10}$$

Where, z_i is the raw output (logit) for class i and e^{z_i} is the exponential of the logit z_i. The denominator is the summation of the exponentials of all logits, this ensures that the output probabilities sum to 1.

After stacking the expert outputs into a tensor, the top-k function from TensorFlow is used to choose the top-k experts (2 experts, default) based on the gating probabilities. Each expert's contribution is scaled by its corresponding gating probability, and the final result is calculated as a weighted sum of the top expert outputs. This makes it possible for the model to concentrate on the experts who are most pertinent to each input, facilitating more specialized and effective decision-making. The original input is then transformed into the weighted combination of the top expert outputs by wrapping the MoE layer in a Tensorflow/Keras Model. The model's ability to adaptively select experts through the MoE architecture improves performance on tasks involving complex decision-making.

5.5. Our final model — ExpressNet-MoE

The CNNFE1, CNNFE2, ResNet-50 and MoE modules are combined in the final ExpressNet-MoE Model (Fig. 7) to produce a robust architecture for emotion recognition.

The model starts with a Keras Input layer that can handle 224 × 224 RGB images. In order to capture varying amounts of picture information, three different CNN-based feature extractors are used: CNNFE1, CNNFE2, and ResNet50. Sequentially putting the input through these models yields their results. Following the concatenation of the CNNFE2 and ResNet50 features, a dense layer with 512 units and ReLU activation is applied to the combined features. A Dropout layer with a regularization rate of 0.5 is then applied.

The two sets of features are subjected to the MoE mechanism: a MoE layer specifically designed for CNN-based features processes the features from CNNFE1, and a second MoE layer also designed for the CNN features processes the dense layer output (from CNNFE2 and ResNet50). The high-level extracted representations of the input image are represented by a composite feature vector created by concatenating the outputs from the two MoE layers which we will call the final features.

To predict one of the emotions in input images, an output layer with softmax activation is added at the end taking the output as an input from the final features. To enhance generalization, the model is assembled using the Adam optimizer and categorical cross-entropy loss with label smoothing. The chosen loss function, categorical Cross Entropy (CCE) is calculated using Eq. (11). In the equation, K is the number of classes, y_i is the true label and $\hat{y}_i$ is the predicted label. $\log(\hat{y}_i)$ is the natural logarithm of the predicted probability of the class i.

$$\text{CCE}(y, \hat{y}) = -\sum_{i=1}^{K} y_i * \log \hat{y}_i \tag{11}$$

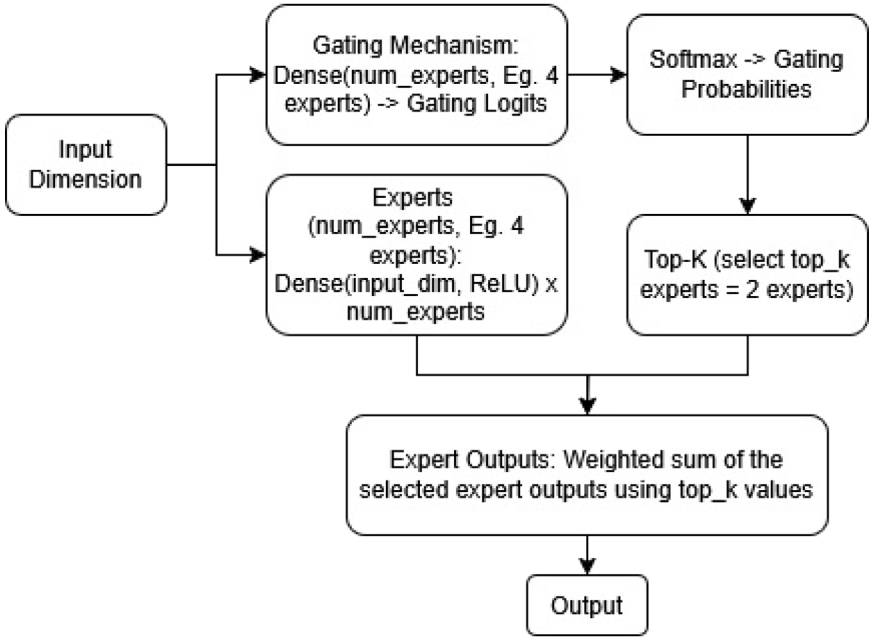

Fig. 6. Mixture of Experts (MoE).

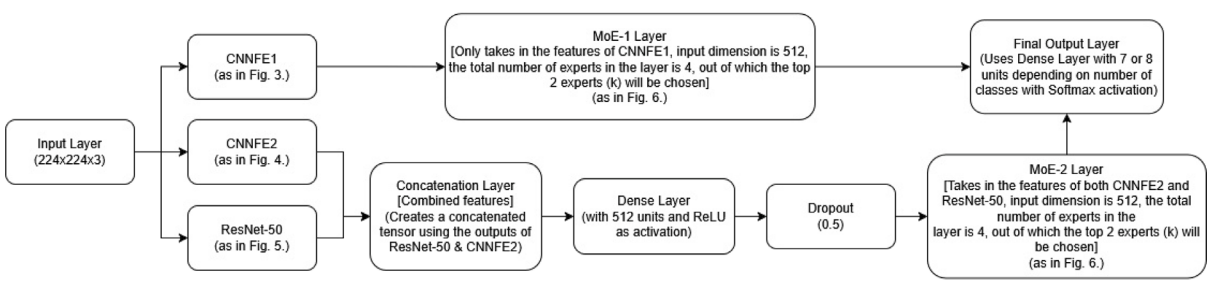

Fig. 7. ExpressNet-MoE architecture.

Table 1
List of hyperparameter values.

Hyperparameter	Value
Neural Network type	Hybrid CNN-based Mixture of Experts
Classification type	Multi-class (7, and 8 classes)
Batch size	32
Dropout rate	Variable (0.1-0.5)
Number of epochs	15
Optimizer	Adam Learning rate, $\alpha = 10^{-4}$ $\beta_1 = 0.9, \beta_2 = 0.999$
Loss function	Categorical Cross-entropy

We ran our experiments in a Tensorflow 2.0 compute framework configured with four NVidia Titan Xp GPU co-processors with mirrored strategy enabled. Values of the hyperparameters were chosen by heuristics and are listed in Table 1.

6. Results

Table 2 lists the classification report of our proposed ExpressNet-MoE model on all of the four datasets in terms of precision, recall, F_1, and per-class accuracy (p-Acc), while Table 3 lists the overall accuracy, macro averaged precision, recall and F_1-score of our model at the evaluation.

The model's performance varies on different datasets, which reflects issues specific to each dataset. Happiness regularly has the highest F_1-scores, showing a good recognition of positive emotions, especially in RAF-DB (0.93) and FER-2013 (0.86). In FER-2013, anger and fear had lower F_1-scores (0.59 and 0.40, respectively), due to class imbalance and incorrect classification. In contrast to AN-8 (0.70), disgust performs badly in FER-2013 ($F_1 = 0.49$), which is directly correlated to having less training and testing data. In FER-2013, neutral expressions have a high recall (0.75) but low precision (0.50), which compromises the reliability of classification. Overall, our model shows consistent performance on all of RAF-DB emotion classes (84.29%) (Table 3), yet it shows similar on prediction accuracies on the two versions of AffectNet datasets: AffectNet$_7$ (74.77%) and AffectNet$_8$ (72.55%).

The generalized trend noticed during the training on all datasets is that validation accuracy typically followed an increasing trend with some fluctuations, indicating possible overfitting, training accuracy demonstrated consistent improvements. Our check-points made sure only the model with the highest validation accuracy was saved to mitigate the overfitting. Adjusting the learning rate facilitated smooth convergence, while early stopping avoided needless training.

6.1. AffectNet$_7$ results

The classification task on AffectNet$_7$ showed a robust learning progression. From 49.51% in the first epoch to 97.60% in the tenth (in Fig. 8). The training accuracy increased significantly. By the sixth epoch validation accuracy had reached 74.86%, followed by slight

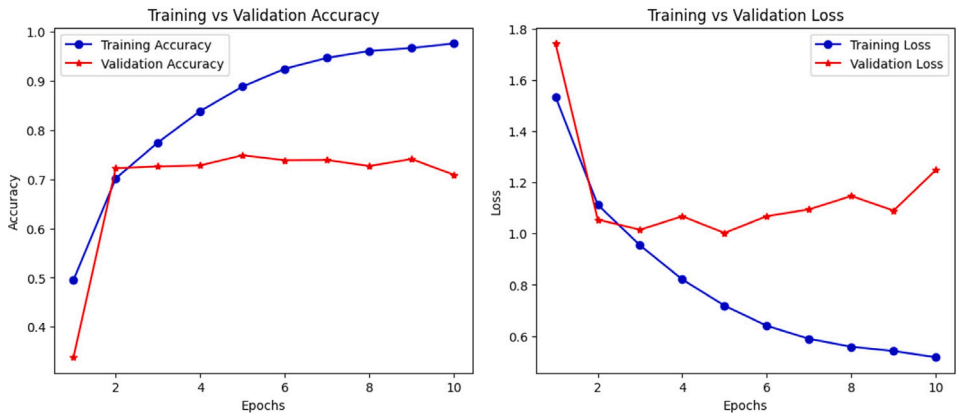

Fig. 8. Training accuracy/loss vs Validation accuracy/loss: AffectNet$_7$.

Table 2
Classification Report of proposed ExpressNet-MoE model on all the test-sets from AffectNet$_7$ (AN-7), AffectNet$_8$ (AN-8), RAF-DB (RAF), FER-2013 (FER).

Emotion	Dataset	Precision	Recall	F_1	Support	p-Acc
Anger	AN-7	0.71	0.79	0.75	722	92.38
	AN-8	0.83	0.55	0.66	722	92.80
	RAF	0.71	0.87	0.78	162	**97.43**
	FER	0.61	0.58	0.59	958	89.40
Disgust	AN-7	0.79	0.52	0.63	695	91.46
	AN-8	0.68	0.72	0.70	694	92.48
	RAF	0.62	0.60	0.61	160	96.02
	FER	0.75	0.36	0.49	111	**98.83**
Fear	AN-7	0.70	0.76	0.73	609	93.18
	AN-8	0.90	0.59	0.71	609	94.85
	RAF	0.65	0.62	0.63	74	**98.27**
	FER	0.65	0.29	0.40	1024	87.66
Happy	AN-7	0.80	0.97	0.88	867	95.31
	AN-8	0.89	0.84	0.86	867	**95.94**
	RAF	0.91	0.95	0.93	1185	94.43
	FER	0.88	0.84	0.86	1774	93.22
Neutral	AN-7	0.70	0.72	0.71	572	**93.18**
	AN-8	0.55	0.73	0.62	572	91.14
	RAF	0.88	0.73	0.80	680	91.75
	FER	0.50	0.75	0.60	1233	82.84
Sad	AN-7	0.76	0.62	0.68	599	93.06
	AN-8	0.62	0.75	0.68	599	92.56
	RAF	0.77	0.88	0.82	478	**93.97**
	FER	0.50	0.64	0.56	1247	82.77
Surprise	AN-7	0.75	0.76	0.76	923	90.98
	AN-8	0.74	0.80	0.77	923	92.00
	RAF	0.87	0.82	0.84	329	**96.71**
	FER	0.85	0.64	0.73	831	94.61
Contempt	AN-8	0.69	0.76	0.72	649	**93.33**

Table 3
Overall evaluation of ExpressNet-MoE model on the test-sets in terms of accuracy and macro averaged Precision, Recall and F_1-score.

Dataset	AffectNet$_7$	AffectNet$_8$	RAF-DB	FER-2013
Accuracy	74.77%	72.55%	84.29%	64.66%
Macro-Prec.	74.56%	73.69%	77.28%	67.86%
Macro-Rec.	73.58%	71.83%	78.00%	58.62%
Macro-F_1	73.41%	71.75%	77.34%	6058%

variations that might indicate overfitting. Despite these differences, the final stored model matches an optimally generalized version that successfully captures emotional patterns in the AffectNet$_7$ dataset. The final test accuracy the model achieved in this dataset is 74.77%.

6.2. *AffectNet$_8$ results*

Training accuracy increased significantly in the AffectNet$_8$ classification task, rising from 47.68% in the first epoch to 96.88% by the ninth (in Fig. 9).

While training accuracy kept increasing, suggesting possible overfitting, validation accuracy peaked at 72.35% in the eighth epoch and then varied slightly. However, optimal generalization is ensured by the final stored model as it stores the model when highest validation accuracy is achieved. Even though the model was able to learn emotion representations, its resilience could be further increased by using additional regularization techniques. The final testing accuracy that the model achieved in this dataset is 72.55%.

6.3. *RAF-DB results*

Training accuracy increased from 58.89% in the first epoch to 97.54% by the ninth (in Fig. 10), demonstrating a rapid improvement on the RAF-DB dataset.

Similar trends were seen in validation accuracy, which first increased from 22.24% to 84.21% in the ninth epoch. The model eventually attained high generalization performance, despite a few slight decreases in the sixth and seventh epochs. Early stopping strengthened the model's stability for practical FER applications by preventing needless training past the point of peak performance. The final testing accuracy that the model achieved in this dataset is 84.29%.

6.4. *FER-2013 results*

A peak validation accuracy of 64.37% and a training accuracy of 93.00% are achieved on the FER-2013 training data (Fig. 11). However, after peaking, validation accuracy fluctuated, mostly as a result of incorrectly categorized data in the dataset and overfitting. Despite this, the model was able to acquire strong emotion representations. Better preprocessing or label correction could improve the results even though the final stored model is the most generalizable version. The final testing accuracy that the model achieved in this dataset is 64.66

In every dataset, the "emotion" happy is classified with the highest accuracy. This is due to the fact that all of the datasets had "happy" images predominantly.

7. Comparative analysis

Table 4 presents a detailed comparison of the ExpressNET-MoE model against other existing models on the AffectNet$_7$, AffectNet$_8$, RAF-DB, and FER-2013 datasets. All baseline results reported in Table 4 are taken directly from the original referenced papers.

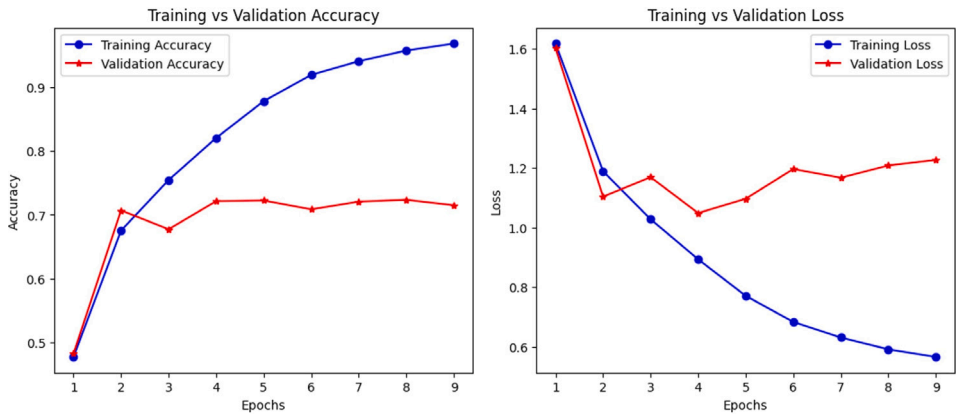

Fig. 9. Training accuracy/Loss vs Validation accuracy/Loss: AffectNet_8.

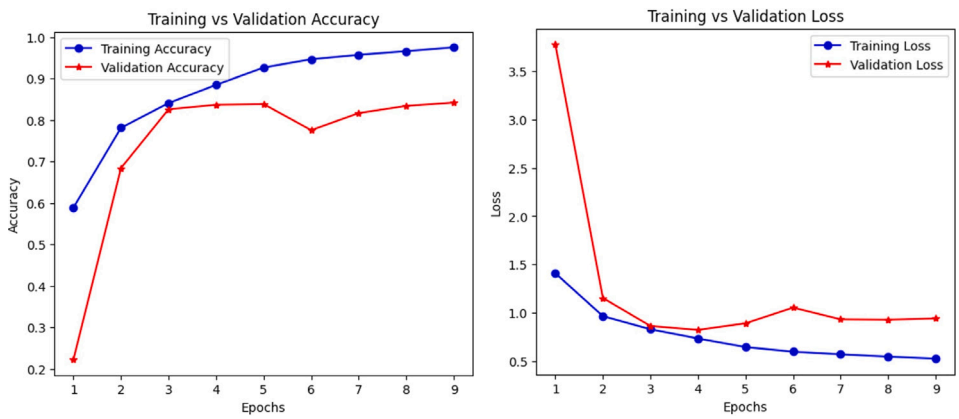

Fig. 10. Training accuracy/Loss vs Validation accuracy/Loss: RAF-DB.

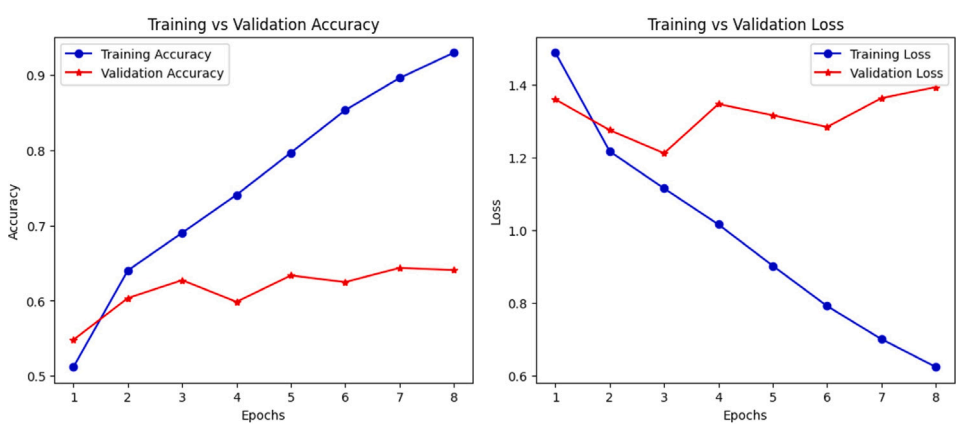

Fig. 11. Training accuracy/Loss vs Validation accuracy/Loss: FER-2013.

We repeated the experiments (with $n = 3$) with our proposed ExpressNet-MoE model, each time by varying the training and test splits, and presented in Table 4, the classification accuracies with mean, standard deviation with p-values (two-sided test) for each of the four datasets that demonstrates there exists no bias due to the variances in the splits.

Our model outperforms all other models as seen in Table 4 as it achieves mean accuracy of 74.40% with standard deviation, $\sigma = 0.45$ on the AffectNet_7 dataset. The models that are its next competitors are ResEmoteNet (72.93%), QCS (67.94%) and EmoNeXt-XL (67.46%), lag behind. Other models fall short too, including EfficientNet-B2 with an accuracy of 66.29% and DDAMFN with an accuracy of 67.03%. This substantial improvement in testing accuracy in our model demonstrates the effectiveness of our model in capturing nuanced facial emotions and fluctuations in AffectNet_7 which is a dataset renowned for its challenging real-world photographs. When compared to single-stream architectures such as EfficientNet and EmoNeXt, our methodology which incorporates multi-expert learning and employs ensemble CNN and ResNet-based feature extraction, has provided a superior representational capacity, which is evidenced by the substantial gap between our model and the preceding best-performing models.

Similarly, our model outperforms all prior approaches in classifying AffectNet_8 dataset by achieving on an average 71.98% (with $\sigma = 0.66$) as its testing accuracy on the dataset. For this dataset, D-CNN has

Table 4
Comparison against SOTA, reported in terms of overall classification accuracy ($AffectNet_7$: AN-7, $AffectNet_8$: AN-8, RAF-DB: RAF, FER-2013: FER).

Model	AN-7	AN-8	RAF	FER
ViT (Roka & Rawat, 2023)	64.48%	–	–	–
MobileNet-v1 (Savchenko et al., 2022)	64.71%	60.25%	–	–
EfficientNet-B0 (Savchenko et al., 2022)	65.74%	61.32%	–	–
EmoNeXt-T (Bohi et al., 2024)	65.55%	61.36%	–	73.34%
EmoNeXt-S (Bohi et al., 2024)	65.90%	62.51%	–	74.33%
EmoNeXt-B (Bohi et al., 2024)	66.21%	62.94%	–	74.91%
EfficientNet-B2 (Savchenko et al., 2022)	66.34%	63.03%	–	–
Face2Nodes (Jiang et al., 2023)	66.69%	–	91.02%	–
EmoNeXt-L (Bohi et al., 2024)	66.88%	63.12%	–	75.57%
EmoNeXt-XL (Bohi et al., 2024)	67.46%	64.13%	–	76.12%
DDAMFN (Zhang et al., 2023)	67.03%	64.25%	91.35%	–
QCS (Wang et al., 2024)	67.94%	64.30%	92.50%	–
DCNN-CBAM (Halim et al., 2023)	–	66.09%	–	72.28%
Norface (Liu et al., 2024)	–	68.69%	92.97%	–
D-CNN (Uniyal & Agarwal, 2024)	–	70%	–	–
D-CNN (Huang, Chiang, et al., 2023)	56.54%	–	83.37%	–
GZS-ConvNet (Bhati et al., 2025)	59.61%	–	62.51%	75.32%
FER-VT (Huang et al., 2021)	–	–	88.26%	–
ResEmoteNet (Roy et al., 2024)	72.93%	–	94.76%	79.79%
ExpressNet-MoE (Ours)	74.40 ± 0.45, ($p \approx 0.0077$)	71.98 ± 0.66, ($p \approx 0.0097$)	83.41 ± 1.06, ($p \approx 0.00016$)	67.05 ± 2.08, ($p \approx 0.0011$)

the highest reported accuracy of 70%, followed by Norface achieving 68.69% and then QCS achieving 64.30%. The margin of 2.55% over D-CNN and 3.86% over Norface underscores our model's effectiveness in generalizing even when an additional emotion class (contempt) is included, which frequently complicates classification tasks due to its subtlety. Fine-tuned EfficientNet-B2 with an accuracy of 63.03% and EmoNeXt-XL with an accuracy of 64.13% perform significantly worse, emphasizing the advantages of our model's architecture for learning intricate facial emotions. The improvement in accuracy between $AffectNet_7$ and $AffectNet_8$ indicates that our model is both resilient to variations in the distribution of emotional classes and capable of managing large-scale datasets.

Our model achieves an average of 83.41% accuracy (with $\sigma = 1.06$) on the RAF-DB dataset which is competitive and proves the model's adaptability and generalizability but still falls short of the top-performing techniques. ResEmoteNet achieves the highest accuracy of an accuracy of 94.76%, followed closely by Norface with an accuracy of 92.97% and QCS with 92.50% accuracy. The models outperform our approach by approximately 8.68% to 10.47%. This can be attributed to the fact that RAF-DB is predominantly imbalanced and frequently requires fine-grained feature extraction. To enhance our ability to extract substantial representations from RAF-DB photographs, supplementary pretraining strategies along with data augmentation would be necessary.

Despite its superior performances on $AffectNet_7$, $AffectNet_8$ and RAF-DB, our model's FER-2013's testing accuracy of 67.05% $\pm$ 2.08 struggled to that of the best-performing baseline models. ResEmoteNet with an accuracy of 79.79%, EmoNeXt-XL with 76.12% accuracy, and GZS-ConvNet with an accuracy of 75.32% all outperformed our method. FER-2013 presents significant challenges due to its grayscale nature and low-resolution photographs (48 × 48 pixels). Additionally, the dataset's mislabeling introduces substantial errors. To enhance performance on FER-2013, state-of-the-art algorithms often employ additional preprocessing methods, transfer learning, or domain adaptation. While our method demonstrates effectiveness on AffectNet, it may not be as well-suited for grayscale photographs, potentially contributing to the observed performance decline. Furthermore, the high amount of noise and label ambiguity in FER-2013 may have an impact on models that depend on high-resolution features. It is worth noting that the majority of models achieving higher accuracies on FER-2013 exhibit poor performance on AffectNet due to presence of many mislabeled samples in it.

8. Discussion

Generalization and adaptability have been enhanced by our proposed model with the dynamic selection of the relevant experts, a capability enabled by the model's mixture of experts (MoE) architecture. Furthermore, the transfer learning method and ensemble CNN architecture also adds to reducing the problems related to applying FER in real-world scenarios due to changes in light conditions, occlusion, and demographic diversity. ExpressNet-MoE provides a robust means for FER applications in real-world scenarios, is capable of recognizing complex emotional expressions and provides a strong base for engagement detection systems as well by classifying emotional states more accurately.

Another important finding of this research is also that the model's capacity to identify some emotions, like happiness, is noticeably more accurate than other emotions like fear or disgust. This is attributed to the datasets' class imbalances, where certain emotions are over-represented while others have few training samples. Even though the model benefits from adaptive feature learning, further improvements can be made involving more complex data augmentation methods or fine-tuning methods using domain-specific datasets. This could increase the recognition of underrepresented emotions by FER models. Additionally, the performance difference between RAF-DB and high-performing models like ResEmoteNet indicates that fine-grained feature extraction may benefit from additional optimization. To improve the model's sensitivity to minute face changes, future research could investigate architectural changes such as adding attention processes.

Furthermore, as we can see in Fig. 12, the distribution of RGB values for AffectNet and RAF-DB are different with RAF-DB having higher pixel values for the red channel. This is attributed to variations in head position, occlusion, lighting conditions and skin color. The difference in the accuracies received for both RAF-DB and AffectNet can be attributed to this aspect of the datasets. We can also see in Fig. 13 that the Mediapipe library that we used for image pre-processing crops the entire face for many RAF-DB images as these images for emotions like surprise, fear, etc. have their faces covered. Therefore, the proposed architecture cannot properly classify these images for the RAF-DB dataset and underperforms for the dataset.

9. Conclusion

To enhance facial emotion recognition, the proposed ExpressNet-MoE presents a novel hybrid deep learning architecture that successfully combines Mixture of Experts (MoE) module with Convolution

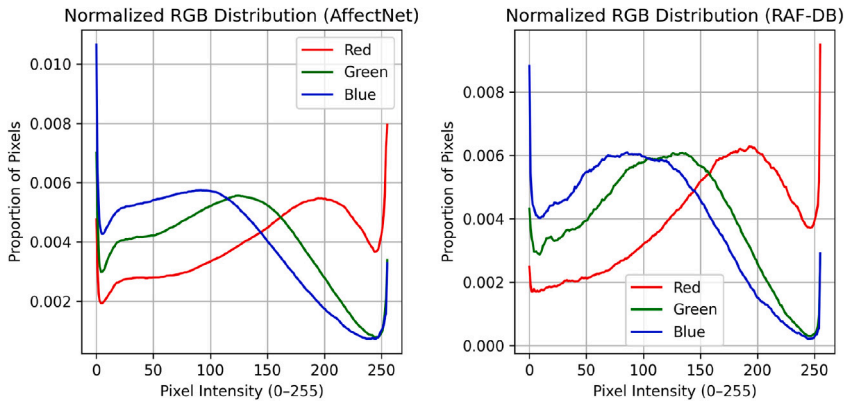

Fig. 12. AffectNet (left) and RAF-DB (right) normalized color distribution across all images.

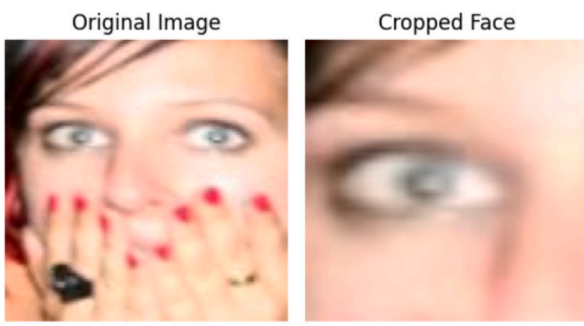

Fig. 13. Mediapipe's detection on RAF-DB.

Neural Networks and provides new insights into how expert models can be dynamically selected for emotion recognition tasks. This advances our knowledge of how ensemble approaches can be used to solve visual recognition problems, especially ones that involve intricate and subtle patterns like facial expressions. The model maintains competitive performance on RAF-DB while achieving excellent accuracy on AffectNet_7 and AffectNet_8 by utilizing dynamic expert selection and multi-scale feature extraction. The findings demonstrate the value of adaptive feature learning in addressing practical facial emotion recognition challenges including occlusion, lighting conditions, and demographic diversity of the subjects. ExpressNet-MoE lays a solid basis for future developments in such systems, despite certain limitations, especially in datasets with noisy labels or unbalanced class distributions. To improve the model's generalizability further in a range of applications, future research could concentrate on honing expert selection techniques or increasing the number of experts, improving image pre-processing as we see improper pre-processing degrades the model's performance, adding attention-based mechanisms, and increasing variance in training datasets.

CRediT authorship contribution statement

Deeptimaan Banerjee: Conceptualization of this study, Methodology, Software, Visualization, Writing – original draft, review and editing. **Prateek Gothwal:** Reviewing and Editing, Validation. **Ashis Kumer Biswas:** Writing – review & editing, Methodology, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ashis Kumer Biswas reports financial support was provided by National Science Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

I have shared publicly accessible link to our code repository. Data used is publicly accessible and links are also provided in the paper.

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Deeptimaan Banerjee received his B.Tech. in Electronics and Computer Engineering from Vellore Institute of Technology, Chennai, India, in 2023. He is currently pursuing his master's in computer science at the University of Colorado, Denver, US. He serves as a Research Assistant and a Teaching Assistant for courses including Deep Learning, Machine Learning, AI with Reinforcement Learning, and Social Informatics & Graph Theory. His research interests include Transfer Learning, Convolution Neural Networks and Transformer-based Networks.

Prateek Gothwal received the B.E. degree in Computer Engineering from Netaji Subhas University of Technology, Dwarka, India. He is currently pursuing the M.Sc. degree in Computer Science at the University of Colorado Denver, CO, USA. He is working as a Research Assistant on student engagement prediction in online classes, focusing on video classification using Vision Transformers. His research interests include deep learning and computer vision.

Ashis Kumer Biswas is an associate professor of the Department of Computer Science and Engineering, and Director of the Machine Learning laboratory (ML Lab) at the University of Colorado Denver. He received his Ph.D. degree in Computer Science from the University of Texas at Arlington, and an M.S. and a B.S. degree in Computer Science and Engineering from University of Dhaka, Bangladesh. Dr. Biswas's research interests span several areas related to responsible machine learning and deep learning algorithm design, implementation that entails considering ethics, bias in the models, privacy-aware model training and energy efficient modeling pipeline.